

Machine Learning for Named Entity Recognition and Classification Report, Theoretical Component, and NERC-Research Preparation

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Abstract

Fill in a short abstract for your final submission.

We recommend aiming for a report of around 6. This does not include acknowledgements or references (they may appear on page 7 and later if needed). You can use appendices if you want to share more details. Below, we provide length indications per section. Please try to stick to them, but don't worry about receiving a lower grade if you don't quite manage. They are supposed to help you and prevent you from doing extra work.

1 Introduction

Length: $\pm 1/2$ - max 1 column for Introduction

Write a short introduction to your approach for the final submission. The introduction should include:

- A brief description of the task (not detailed)
- An outline of the ML approaches included in the report
- A brief summary of your results

Assignment 3

2 Task and Data

Length: ± 1.5 -2 columns for Task and Data.

2.1 Task

Assignment 1 NER task description

Assignment 2 Update if necessary

Assignment 3 Update if necessary

2.2 Dataset and Distribution

Assignment 1 Data description and distribution

Assignment 2 update if necessary

Assignment 3 update if necessary

2.3 Preprocessing

Assignment 1 Preprocessing you did for first analysis

Assignment 2 update (if applicable)

Assignment 3 update (if applicable)

2.4 Evaluation Metrics

Assignment 1

Assignment 2 and 3: Update if necessary

3 Models and Features

Length: ± 2 columns for Models and Features.

3.1 Models

Assignment 1 Logistic Regression

Assignment 2 Alternative Methods (SVM, NB)

Assignment 3 More advanced models (fine-tuning BERT, CRF for ReMA students only)

3.2 Features

Assignment 1 Feature exploration

Assignment 2 More features

Assignment 3 More advanced features

4 Experiments and Results

Length: ± 3 columns (including figures and tables) for Experiments and Results. Additional material can be provided in the Appendix.

4.1 Evaluation

Assignment 1: first results

Assignment 2: update

Assignment 3: update

4.2 Feature Ablation

Assignment 3: feature ablation for one system

Note: if you decide after your feature ablation that you would like to discuss this before the Evaluation, you may move this subsection.

5 Error Analysis

Length: ± 1.5 -2 columns for Error Analysis.

Assignment 3: beyond the confusion matrix

6 Discussion

Length: ± 1 column for Discussion and Conclusion together.

Assignment 3

7 Conclusion

Assignment 3

Acknowledgements

References

Ethem Alpaydin.2020. *Introduction to Machine Learning Introduction to Machine Learning*, 4 edition. MIT Press.

Tom M Mitchell. 1997. *Machine Learning*. McGraw-Hill.

Theoretical Component

Assignment 1 - What is Machine learning?

My answer on the question "What is Machine learning" is based on the name itself. Machine learning is about machines learning, but this sounds still very silly and unclear. Therefore, let's first dive into the machines part. People should not think about machines like the first typewriter, or motor engines, or even your normal kitchen blender. We should also not think about the computers or calculators that run simple algorithms. Algorithms are mostly strictly defined, like the ones embedded in your regular calculators. These machines do not learn, they just follow a set of rules that are already defined. This is not machine learning, it is following a recipe. If the recipe is bad, you will get a burned cake. If you don't adjust the oven temperature next time, the cake will burn over and over again. The machines we are looking at are different. They can learn from data, identify patterns, and make decisions based on that data ([Mitchell, 1997](#)). To get back to our cake, this means that after one time burning the cake, it will lower the temperature next time, or even adjust the baking time. It can be undercooked that time, but it will learn from that as well, and adjust again. This learning from data and identifying patterns is the essence of machine learning ([Alpaydin, 2020](#)).

Assignment 2

Assignment 3

Time spent

Week	Task	Time
1	understand the assignment and getting everything setup	30 minutes
1	watch videos	30 minutes
1	understand labels in data	1 hour
Total	2h	

Table 1: Time overview.

A Example Appendix

This is an appendix. You can include additional analyses and tables here (e.g. samples you analyzed for your error analysis).

NERC-research preparation

2.1.1 - NER given the CoNLL dataset

2.2.1 - How does CoNLL work?